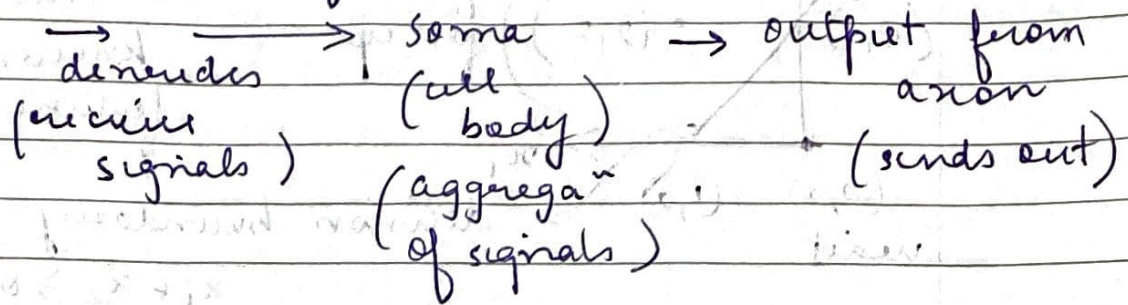


Q1 - 20
Q2 - 20
Addig - 20 /
Pgs - 40

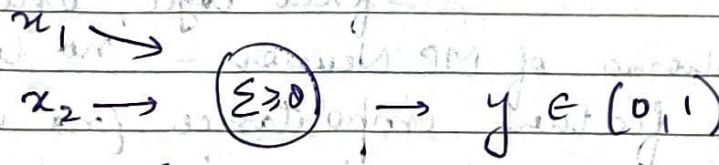
21st Jan 2022 / DA6401

→ From bio to Maths

- Neuron (useful abstraction, not perfect)



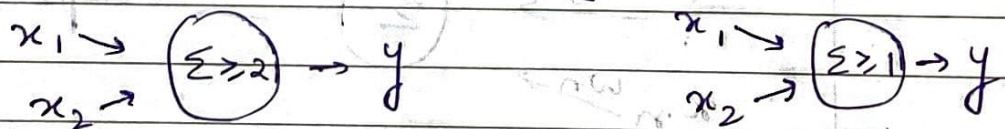
- MP Neuron



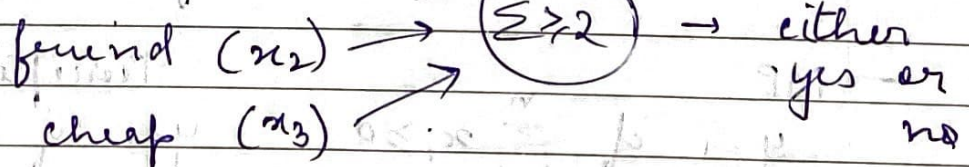
x_i are binary y is binary

y will be 1, if the sum is greater than or equal to the threshold:

It was able to compute any logical units (AND, OR)



- the real-world analogy → movie go or not comedy (x_1)



$x_i \rightarrow$ features

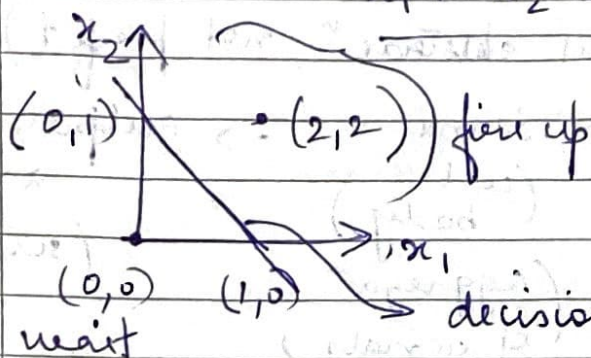
for yes/no _/_/_ answers

- mathematical intuition for 2 inputs

linear classifier

$$x_1 + x_2 \geq 0$$

(MP model)



this space is known as feature space.

$$x_1 + x_2 \geq 0$$

$$x_1 + x_2 = 0$$

linearly separable.

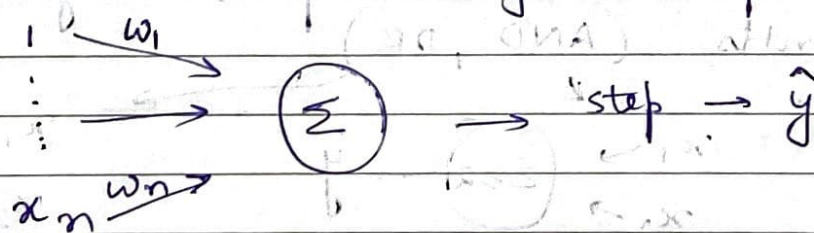
fixed logic, only computes,

- limitations of MP Neuron - no learning

1. no feature importance (no weights)
2. no automatic learning
3. restricted to binary 1/0

⇒ Perceptron (1958)

weighted inputs



learning rule → compares $\hat{y}$ with y .

new machine could classify patterns & it could automatically learn

$$z = w^T x + b$$

- MP

Perceptron

$$y = 1 \text{ if } \sum_{i=1}^n x_i \geq 0$$

$$y = 1 \text{ if } \sum_{i=1}^n w_i x_i + b \geq 0$$

introduces non-linearity

—/—/—

- in the perceptron, we can assign wts. based on the importance.
- once we go into n pts, instead of lines, we get hyperplanes.
- the weight vector is always orthogonal ($\perp$) to the decision boundary. It always direct towards 'yes' class.